Table 1: Raw mass and length measurements of the string segment

Mass m (g)	Length L (cm)
0.0286	100.00
0.0338	115.00
0.0299	100.00

Table 2: Raw standing-wave resonance measurements at different hanging masses

Hanging mass m (g)	Antinode count n	Resonance frequency f_n (Hz)	Antinode width $\lambda_n/2$ (cm)
50.26	2	100.5	62.5
50.26	3	150.5	42.3
50.26	4	201.2	32.0
60.64	2	111.6	61.5
60.64	3	167.9	42.7
60.64	4	223.5	31.9
70.06	2	118.8	63.5
70.06	3	178.7	41.7
70.06	4	238.8	33.0
80.16	2	127.0	63.3
80.16	3	191.2	44.2
80.16	4	255.2	31.0
90.10	2	135.2	63.5
90.10	3	204.2	40.5
90.10	4	272.2	33.1

Table 3: Wave speed inferred from the resonance measurements, at different rope tensions

Rope tension $T = mg$ (N)	Wave speed $v = f_n \lambda_n$ ($\mathrm{m\,s^{-1}}$)
0.4925	127.24 ± 0.91
0.5943	141.1 ± 2.0
0.6866	152.5 ± 2.6
0.7856	162.7 ± 3.3
0.8830	172.4 ± 4.3

Table 4: Regression of v^2 against T, and the resulting estimate of the string's linear density

Quantity	Value
Slope s ($\mathrm{m^2\,s^{-2}\,N^{-1}}$)	$(3.54 \pm 0.23) \times 10^4$
$\mu_{\text{obs}} \equiv 1/s$ ($\mathrm{kg\,m^{-1}}$)	$(2.82 \pm 0.19) \times 10^{-5}$
μ_{ref} ($\mathrm{kg\,m^{-1}}$)	$(2.930 \pm 0.038) \times 10^{-5}$
Error	-3.63%

Table 5: Raw resonance frequency measurements of the metal ring at different antinode counts

Antinode count n	Resonance frequency f (Hz)
3	20.5
4	44.2
5	73.6
7	153.2
9	266.4
11	398.9

Table 6: Regression of $\log f$ against $\log n$ for the metal ring, compared with the theoretical exponent of 2

Quantity	Value
Slope s	2.333 ± 0.061
Error	16.64%

Table 7: Comparing the observed resonance frequency with the f-n^2 fit's prediction, at antinode counts not used in the fit

Antinode count n	Observed frequency f_{obs} (Hz)	Predicted frequency f_{theo} (Hz)	Error
4	44.2	43.1	2.55%
11	398.9	401.5	−0.64%

Table 8: Basic measurements of the spring used in the longitudinal-wave experiment

Quantity	Value
Natural length L_0 (cm)	7.50
Spring mass M (g)	11.46

Table 9: Raw scale readings used to measure the spring constant

Hanging mass m (g)	Upper-loop scale reading L_{up} (cm)	Lower-loop scale reading L_{low} (cm)
10.14	62.30	54.10
20.04	62.05	46.50
30.14	61.90	38.05
40.06	61.90	29.85
50.30	61.85	21.50

Table 10: Extension of the spring under different restoring forces

Restoring force $F = mg$ (N)	Extension $x = L_{\text{up}} - L_{\text{low}} - L_0$ (m)
0.09937	0.0070
0.1964	0.0805
0.2954	0.1635
0.3926	0.2455
0.4929	0.3285

Table 11: Regression of F against x; by Hooke's law the slope is the spring constant K

Quantity	Value
Slope K ($\mathrm{N\,m^{-1}}$)	1.216 ± 0.014
y-intercept (N)	0.0946 ± 0.0027

Table 12: Raw resonance frequency measurements of the spring's longitudinal standing wave

Antinode count n	Resonance frequency f (Hz)
7	36.8
8	41.9
9	47.1
10	52.5
11	57.5

Table 13: Regression of f against n for the spring, compared with the theoretical slope $\sqrt{K/M}/2$

Quantity	Value
Slope s (Hz)	5.200 ± 0.033
Theoretical slope $s_{\text{theo}} = \sqrt{K/M}/2$ (Hz)	5.151 ± 0.029
Error	0.94%

Table 14: Regression of $\log f$ against $\log n$ for the spring, compared with the theoretical exponent of 1

Quantity	Value
Slope s	0.9919 ± 0.0060
Error	-0.81%
y-intercept b	0.7271 ± 0.0057
Theoretical y-intercept $b_{\text{theo}} = \log s_{\text{theo}}$	0.7119 ± 0.0024
Error	2.14%